

How Blockchains Help the Healthcare Industry

The National Institute of Standards and Technology (NIST) defines blockchain as “tamper evident and tamper resistant digital ledgers implemented in a distributed fashion (i.e., without a central repository), usually without a central authority like a bank, company or government. At its basic level, it enables a community of users to record transactions in a shared ledger within that community, such that under normal operation of the blockchain network no transaction can be changed once published.” This article lists the benefits blockchain brings to the healthcare industry.



Blockchain comprises blocks linked or chained together by cryptographic signatures, each of which is called a hash, stored in shared ledgers and supported by a network of connected processes called nodes. Nodes maintain a copy of the entire chain, and are continually updated and kept in sync.

Blockchain technology has the potential to transform healthcare, placing the patient at the centre of the ecosystem and increasing the security, privacy and interoperability

of health data. This technology could provide a new model for health information exchange (HIE) by making electronic medical records more efficient and secure. Blockchain applications in the healthcare industry span from health data exchange and identity management to the drug supply chain, and from insurance to personalised medicine. The European healthcare ecosystem, for example, faces difficult challenges, such as an ageing population, multi-morbidity, healthcare workforce shortages and the rising burden of preventable and non-

communicable diseases. Therefore, it is important to explore blockchain as a new way to organise healthcare services and develop more resilient, accessible and effective quality care for European citizens.

Blockchain has many unique characteristics, three of which make it very useful for the healthcare and medical supply chain industry:

- Full traceability of any information on the blockchain
- Ability to ensure data has not been tampered with
- Enhanced security